

# **CYBER THREAT ANALYTICS DASHBOARD**

Power BI Project Documentation

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# 1. Objective

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The Cyber Threat Analytics Dashboard is a Power BI business intelligence solution built to help security operations and management teams monitor, analyze, and anticipate cybersecurity incidents across an organization's locations, departments, and asset base.

The dashboard consolidates incident and response data into a single analytical model and presents it through seven interactive report pages, each targeted at a different stage of the security decision-making process:

- Understanding what is happening right now (security intelligence and volumes)
- Understanding where incidents are occurring (geospatial and city-level detail)
- Understanding when incidents occur (temporal patterns by hour, day, shift, and season)
- Understanding how well the organization is responding (SLA, escalation, resolution)
- Understanding where risk is concentrated (structured risk scoring across locations, departments, assets, and threat categories)
- Anticipating what is likely to happen next (14-day forecasting with backtested accuracy)
- Summarizing all of the above for leadership (a single executive view of risk, compliance, and performance)

The purpose is to give both technical analysts and non-technical decision-makers a shared, evidence-based view of cyber risk — replacing static incident reports with an interactive model that supports faster, better-informed decisions.

# 2. Dataset and Data Model

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The dashboard is built on a star-schema data model inside Power BI, consisting of two fact tables and six dimension tables, connected through a dedicated measures table. This structure keeps the model efficient and allows every visual to filter and cross-filter consistently across pages.

## 2.1 Fact Tables

**fact\_incident** — the core incident log. Each row represents a single recorded cyber incident, carrying attributes used throughout the report such as severity level, asset type, weather condition, exposure index, business period, and the hour and shift in which the incident occurred.

**fact\_response** — the operational response log linked to incidents. It captures how each incident was handled: response time, resolution time, resolution status, SLA outcome (met/missed), escalation level, reporting department and role, units deployed, and a calculated vulnerability score.

## 2.2 Dimension Tables

**dim\_date** — a standard date dimension supporting the Date Hierarchy (Year / Quarter / Month / Day), month name, week label, and day of week — the backbone of all time-based trend and forecast visuals.

**dim\_location** — geographic reference data (city, state, region) used for the map visuals, city drillthrough, and location-based risk scoring.

**dim\_severity** — the severity classification of incidents (severity level), used in severity breakdowns and slicers.

**dim\_threat\_type** — the categorisation of threats (threat category), used in threat distribution and monthly trend visuals.

**dim\_organization** — organization type reference data, used in exposure and risk-by-organization visuals.

**dim\_asset** — asset reference data (asset type, asset criticality), used in vulnerability and exposure analysis.

**dim\_source** — data source reference dimension, reflected in the “Sources Connected” KPI on the Security Intelligence page.

## 2.3 Measures Table

A dedicated table named “\_Measures” holds all DAX calculations used across the report. Centralising measures in one table is standard Power BI practice — it keeps calculations independent of any single fact table and makes them easy to reuse across every page and visual.

## 2.4 Model Relationships

The two fact tables connect outward to the shared dimension tables (date, location, severity, threat type, organization, and asset), forming a classic star schema. This design lets a single slicer — such as the City slicer on the Temporal Incident Analytics page — filter every visual on the page simultaneously, because all visuals ultimately trace back to the same conformed dimensions.

## 3. DAX Measures

The model contains 41 DAX measures, organised here by the analytical purpose they serve. Measure names are taken directly from the \_Measures table in the .pbix file; groupings reflect how each measure is actually consumed on the report pages.

### 3.1 Core Volume and Activity Measures

| Measure                            | Purpose                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Total Incidents                    | Base count of recorded incidents; the primary volume metric referenced across nearly every page and visual.                 |
| Critical Incidents                 | Count of incidents flagged at the highest severity level.                                                                   |
| Closed Incidents                   | Count of incidents that have reached a closed/resolved status.                                                              |
| Daily Incident Count               | Incident volume aggregated at a daily grain, used in the SLA-by-department stacked view.                                    |
| Sources Connected                  | Count of distinct data sources feeding the model, shown as a Security Intelligence KPI card.                                |
| Normal Incidents / Spike Threshold | Supporting measures used on the spike-detection line chart to separate typical daily volume from a defined spike threshold. |

### 3.2 Response and SLA Measures

| Measure             | Purpose                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------|
| Avg Response Time   | Average time between incident report and first response.                                |
| Avg Resolution Time | Average time between incident report and final resolution.                              |
| SLA Compliance %    | Percentage of incidents resolved within the agreed service-level target.                |
| SLA Breach Rate     | Inverse of SLA compliance — the share of incidents that missed the SLA target.          |
| SLA Miss Rate       | SLA miss rate calculated at threat-category level for the risk scorecard.               |
| Response Delay      | Delay of actual response time against the SLA target, driving the response-delay gauge. |
| Escalation Rate     | Proportion of incidents that were escalated beyond first-level response.                |
| Critical Rate       | Proportion of total incidents classified as critical.                                   |
| Open Backlog %      | Share of incidents still open / unresolved at the time of reporting.                    |

Table 2: Response-time, SLA, and escalation measures.

### 3.3 Risk Scoring Measures

| Measure                                   | Purpose                                                                                                                                          |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Risk Score                                | General-purpose risk score used in the Geospatial Intelligence and City Details drillthrough pages.                                              |
| Location Risk Score / Location Risk Level | City-level risk score and its categorical banding, used in the risk scorecard and forecasting tables.                                            |
| Regional Risk Score / Regional Risk Band  | Same risk-scoring logic rolled up to region level.                                                                                               |
| Operational Unit Risk Score               | Risk score calculated per reporting department, used in the “Risk by Operational Unit” table.                                                    |
| Weakness Score                            | Composite score combining SLA miss rate, escalation rate, and critical rate per threat category, used in “Security Weakness by Threat Category.” |
| Demand Z / Supply Z / Gap Score           | Standardised (z-score) comparison of incident demand versus response capacity by city, used to flag resource allocation gaps.                    |
| Overall Risk Score                        | Single top-line risk figure summarising the organization's overall exposure, shown on the Executive Intelligence Dashboard.                      |
| Critical Risk Indicators                  | Executive-level count of critical risk conditions.                                                                                               |
| High Risk Locations                       | Count of locations classified as high risk, shown as an executive KPI.                                                                           |
| Emerging Risk Zone / Emerging Risk Zones  | Flag and count identifying locations with a rising incident trend — i.e. areas becoming riskier, not just currently risky.                       |

### 3.4 Predictive and Forecasting Measures

| Measure                                    | Purpose                                                                                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Incident Trend Slope                       | Direction and steepness of each location's incident trend, feeding the Emerging Risk Zone logic.                                              |
| Predicted Risk                             | Forward-looking risk indicator shown on the Executive Intelligence Dashboard.                                                                 |
| Backtest Forecast                          | The model's forecasted incident value for a historical date, used to test forecast accuracy against what actually happened.                   |
| Backtest Absolute Error / Backtest Error % | Difference between the backtested forecast and the actual incident count, in absolute and percentage terms.                                   |
| Backtest MAE 14 Days                       | Mean Absolute Error of the forecast across the 14-day backtest window — the single accuracy figure shown on the Predictive Intelligence page. |

### 3.5 Presentation / Utility Measures

Dashboard Title, Dashboard Title1, and Dashboard Title8 are dynamic title-text measures used to display context-sensitive headers on the Security Intelligence, Geospatial Intelligence, and Security Risk Assessment pages respectively.

## 4. Dashboard Pages — Structure Overview

The report contains seven pages that a user navigates directly, plus one drillthrough page (City Details) and one custom tooltip page (Tooltip) that support them. The table below lists them in the order they appear in the report navigation.

| # | Page                             | Role                                                                                            |
|---|----------------------------------|-------------------------------------------------------------------------------------------------|
| 1 | Security Intelligence            | Landing page — top-line incident volumes, severity mix, and threat categories.                  |
| 2 | Geospatial Intelligence          | Where incidents happen — maps, city/organization/asset breakdowns.                              |
| 3 | Temporal Incident Analytics      | When incidents happen — hour, day, shift, weekday, and monthly patterns.                        |
| 4 | Incident Response Analytics      | How well incidents are handled — SLA, escalation, resolution.                                   |
| 5 | Security Risk Assessment         | Structured risk scoring across organizations, assets, locations, departments, and threat types. |
| 6 | Predictive Intelligence          | 14-day forecasting, key influencers, and backtested forecast accuracy.                          |
| 7 | Executive Intelligence Dashboard | Single-screen summary of risk, compliance, and performance for leadership.                      |
| — | City Details (drillthrough)      | Detail page reached by drilling through from a city on any main page.                           |
| — | Tooltip (custom tooltip)         | Hover tooltip page showing city-level incident detail on chart hover.                           |

## 5. Page 1 — Security Intelligence

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This is the report's landing page and gives an at-a-glance summary of overall incident activity, severity distribution, and threat mix.

### 5.1 KPI Cards

- Total Incidents — overall incident count for the selected filter context.
- Source Connected — number of data sources feeding the model (Sources Connected measure).
- Threat Severity Index (TSI) — a composite severity indicator summarising how serious incidents have been overall.
- SLA Compliance — the percentage of incidents resolved within target.
- Avg Response Time — the average time to first response.
- Critical Incidents — count of the highest-severity incidents.

### 5.2 Supporting Visuals

- A donut chart breaking down Total Incidents by severity level (`dim_severity`).
- A clustered bar chart of Total Incidents by threat category (`dim_threat_type`).
- A line chart of Total Incidents by month, showing the overall incident trend over time.
- A detail table combining region, Total Incidents, Threat Severity Index, Avg Response Time, and SLA Compliance — giving a single regional summary in tabular form.
- A clustered bar chart of Threat Severity Index by weather condition, linking environmental context to threat severity.

Together, these visuals answer the first question any security stakeholder asks: how much is happening, how severe is it, and what kind of threats are involved.

## 6. Page 2 — Geospatial Intelligence

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This page shifts the focus from “what” to “where,” mapping incident activity and risk across geography, organization type, and asset type.

### 6.1 Key Visuals

- A map visual plotting incidents by city (`dim_location.city`).
- A filled (choropleth) map showing incident intensity by state.
- A pivot table of city against Risk Score, giving a sortable location-level risk list.
- A clustered bar chart of total response time by city.
- Bar charts breaking down Total Incidents by organization type (`dim_organization`) and by asset type.
- A bar chart of Total Incidents by reporting department.
- A dynamic “Dashboard Title” card that updates page context text.

An action button on this page links to the City Details drillthrough page, letting a user click any city and open a focused, single-city view.



## 7. Page 3 — Temporal Incident Analytics

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This page analyses when incidents occur — by hour, day, week, weekday, shift, and month — to reveal timing patterns that matter for staffing and monitoring decisions.

### 7.1 Key Visuals

- A column chart of Total Incidents by hour of day (fact\_incident.Hour), highlighting peak activity windows.
- Line charts of Total Incidents by week label and by day (using the Date Hierarchy), showing short- and long-term trend.
- A clustered bar chart of Total Incidents by weather condition.
- A 100%-stacked column chart of Total Incidents by shift, showing how incident share is distributed across work shifts.
- Two donut charts: Total Incidents by day type (weekday/weekend) and Total Incidents by weekday.
- A spike-detection line chart comparing Normal Incidents against a defined Spike Threshold across the full date hierarchy — used to visually flag days where incident volume exceeded the normal range.
- A stacked area chart of incidents by month and threat category, showing how the mix of threats shifts over the year.

### 7.2 Filters on This Page

Three filter controls — a severity-level list, a City slicer, and a month list — let a user narrow the entire page down to a specific severity band, city, or time window without leaving the page.

## 8. Page 4 — Incident Response Analytics

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This page evaluates how effectively the organization responds to incidents once they are reported — covering SLA performance, escalation behaviour, and resolution outcomes.

### 8.1 KPI Cards

- Avg Response Time and Avg Resolution Time — the two core speed-of-response metrics.
- SLA Compliance — percentage of incidents meeting the SLA target.
- Units Deployed — total response units deployed (Sum of fact\_response.units\_deployed).

### 8.2 Supporting Visuals

- A 100%-stacked column chart of SLA Compliance by department, comparing which departments meet SLA targets most consistently.
- A donut chart of Resolution Status Distribution (open, in-progress, closed, etc.).
- A bar chart of Incidents by Escalation Level, showing how far incidents typically escalate.

- A clustered column chart of Response Time by Role, comparing responder roles.
- A gauge visual, Response Delay vs SLA Target, giving an immediate visual read on whether response times are inside or outside target.
- A column chart of Resolution Time Distribution in minutes (binned), showing the spread of resolution durations.
- A donut chart contrasting Critical Incidents against Closed Incidents.

Role, department, and escalation-level list filters allow this page to be narrowed to a specific team or escalation tier for operational review.

## 9. Page 5 — Security Risk Assessment

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This page consolidates risk scoring logic into a single structured view, letting stakeholders see where risk concentrates across six different dimensions of the organization.

### 9.1 Key Visuals

- Exposure Index by Organization Type — a clustered bar chart comparing exposure across organization types.
- Exposure Index by Asset Criticality — comparing exposure across asset-criticality tiers (e.g. low/medium/high/critical assets).
- Vulnerability by Asset Type — a table ranking asset types by Vulnerability Score.
- Location Risk by City — a table combining Location Risk Score and Location Risk Level per city.
- Risk by Operational Unit — a table showing Operational Unit Risk Score by reporting department.
- Resource Allocation Gap by City — a table combining Demand Z, Supply Z, and Gap Score, identifying cities where incident demand outpaces response capacity.
- Security Weakness by Threat Category — a table combining SLA Miss Rate, Escalation Rate, Critical Rate, and Weakness Score per threat category, pinpointing which threat types are handled least effectively.
- Security Risk Scorecard — a table combining Regional Risk Score and Regional Risk Band, rolling risk up to the regional level.

This page is the analytical core of the dashboard: it turns raw incident and response data into structured, comparable risk scores across six different organizational lenses — organization type, asset criticality, asset type, city, department, and threat category — so risk owners can prioritise where to act first.

## 10. Page 6 — Predictive Intelligence

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This page moves the dashboard from describing the past to anticipating the near future, using Power BI's built-in forecasting alongside a rigorous backtest of that forecast's accuracy.

### 10.1 14-Day Forecasting

The primary visual, Incident Trend & 14-Day Forecast, is a line chart of Total Incidents against the date hierarchy with a forecast extended 14 days beyond the last known date. This projects expected incident volume for the next two weeks based on the historical trend, giving security leadership advance notice of likely load rather than only a rear-facing report.

## 10.2 Key Influencers

A Key Influencers visual (“Predictive Threat Risk — Key Influencers”) analyses which factors are most associated with incident severity, using severity level, threat category, weather condition, asset criticality, business period, and hour of day as candidate influencers. This surfaces, in plain language, which conditions most often coincide with more severe incidents — useful both for security investigation and for explaining the forecast to a non-technical audience.

## 10.3 Backtesting the Forecast

Because a forecast is only useful if it can be trusted, this page includes a dedicated backtest of the 14-day forecast against dates where the actual outcome is already known:

- 14-Day Forecast Backtest — a line chart comparing Total Incidents (actual) against the model's forecast for the same historical period.
- Actual vs Forecast — 14-Day Backtest — a second line chart pairing Total Incidents with Backtest Forecast directly, for visual comparison of how closely the forecast tracked reality.
- A supporting table listing, by day, the actual Total Incidents, Backtest Forecast, Backtest Absolute Error, and Backtest Error % — giving day-by-day transparency into forecast accuracy rather than a single opaque number.
- 14-Day Forecast MAE — a KPI card showing the Mean Absolute Error of the forecast across the full 14-day backtest window, giving one clear figure for “how accurate is this forecast, on average.”

This backtest-then-forecast approach is a best practice borrowed from applied forecasting: rather than presenting a forecast as fact, the dashboard first demonstrates, using historical data, how accurate that same forecasting method has been — then applies it forward. This is what makes the 14-day forecast defensible in a technical interview setting: the model's error rate is measured and shown, not assumed.

## 10.4 Emerging Risk Zones

A supporting table on this page combines city, Location Risk Score, Incident Trend Slope, and Emerging Risk Zone — identifying locations where the incident trend is rising even if the location's current absolute risk score is not yet the highest. This distinguishes “already risky” locations from “becoming risky” locations, which is a materially different signal for a security team planning where to add monitoring or headcount next.

# 11. Page 7 — Executive Intelligence Dashboard

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This page condenses the entire model into a single screen of KPI cards, designed for leadership who need the headline numbers without navigating the operational pages.

## 11.1 Executive KPIs

| KPI Card                    | What It Tells Leadership                                                          |
|-----------------------------|-----------------------------------------------------------------------------------|
| Overall Security Risk Score | The single top-line figure summarising organization-wide risk exposure.           |
| High-Risk Locations         | Count of locations currently classified as high risk.                             |
| Critical Incidents          | Count of the most severe incidents recorded.                                      |
| Predicted Risk              | Forward-looking risk indicator drawn from the predictive model.                   |
| Emerging Risk Zones         | Count of locations with a rising incident trend — an early-warning indicator.     |
| SLA Compliance              | Percentage of incidents resolved within target — an operational health indicator. |
| Open Backlog                | Share of incidents still unresolved.                                              |
| Escalation Rate             | Proportion of incidents requiring escalation.                                     |

## 11.2 Strategic Performance Metrics

A second row of cards adds performance-oriented context alongside the risk figures:

- Average Resolution Time and Average Response Time — organization-wide speed benchmarks.
- Critical Incident Rate — critical incidents expressed as a rate rather than a raw count, for trend comparison over time.
- SLA Breach Rate — the inverse view of SLA Compliance, framed as a risk metric.

Two supporting text boxes on this page provide narrative framing for the KPI grid, helping a non-technical viewer interpret the numbers without needing to open the underlying operational pages.

## 11.3 Why This Page Matters for Decision-Making

The Executive Intelligence Dashboard is deliberately built to answer three questions a manager or CISO asks in under a minute: how much risk exists right now, is that risk trending up or down, and is the response organization keeping pace with it (SLA, backlog, escalation). By combining current-state risk (Overall Risk Score, Critical Incidents), forward-looking risk (Predicted Risk, Emerging Risk Zones), and response performance (SLA Compliance, Open Backlog, Escalation Rate) on one page, it supports resourcing and prioritisation decisions without requiring the reader to interpret DAX, charts, or raw data.

# 12. Drill through and Tooltip Pages

## 12.1 City Details (Drillthrough Page)

City Details is configured as a drillthrough page filtered by `dim_location.city`. Right-clicking (or using the action button) on a city anywhere in the report opens this page pre-filtered to that single city, showing:

- Total Incidents, Risk Score, and Critical Incidents for the selected city.
- A pivot table cross-tabulating severity level against Total Incidents for that city.

This gives an analyst a focused, single-location deep dive without needing to manually filter every visual on the main pages.

## 12.2 Tooltip (Custom Tooltip Page)

Tooltip is configured as a Power BI custom tooltip page (page type: tooltip), also filtered by `dim_location.city`. It displays Total Incidents and a supporting bar chart, and appears automatically when a user hovers over a city-related data point elsewhere in the report — giving quick contextual detail without a full page navigation or click.

## 13. Interactivity Features

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The dashboard is built to be explored, not just viewed. Interactivity is delivered through several standard Power BI mechanisms, all present in the model:

- Cross-filtering: because every visual traces back to the shared star-schema dimensions, clicking any category (e.g. a severity level or threat category) filters every other visual on the same page.
- Slicers and filter lists: severity, city, month, department, role, and escalation-level slicers appear on the Temporal Incident Analytics and Incident Response Analytics pages, letting users narrow the view interactively.
- Drillthrough: the City Details page lets a user drill from any city-level visual into a focused single-city report.
- Custom tooltips: the Tooltip page delivers contextual city-level detail on hover, without leaving the current page.
- Action buttons: navigation buttons on the Geospatial Intelligence, City Details, and Predictive Intelligence pages support guided movement between pages.
- Dynamic titles: measures such as Dashboard Title, Dashboard Title1, and Dashboard Title8 update page headers based on the active filter context.

## 14. Key Insights the Dashboard Enables

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Rather than presenting isolated charts, the pages are designed to build on one another, so a user moving through the report can answer a connected sequence of questions:

- Which threat categories and severity levels dominate incident volume (Security Intelligence).
- Which cities, states, and organization/asset types are most exposed (Geospatial Intelligence).
- When incidents are most likely to occur — by hour, shift, weekday, or season — and whether any day has spiked above the normal range (Temporal Incident Analytics).
- How consistently the response organization meets its SLA, and where escalation and resolution times are weakest (Incident Response Analytics).
- Where risk concentrates across six organizational dimensions at once, and which cities show a demand-versus-capacity gap (Security Risk Assessment).
- What the next 14 days of incident volume is likely to look like, how accurate that forecast has historically been, and which locations show a rising (emerging) risk trend even before they become top-ranked by absolute risk (Predictive Intelligence).
- A single-page summary of current risk, predicted risk, and response performance for leadership review (Executive Intelligence Dashboard).

## 15. Recommendations

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Based strictly on what the dashboard already measures and surfaces, the following actions follow directly from its outputs:

- Prioritise remediation in locations and departments flagged with high Location Risk Score / Location Risk Level or high Operational Unit Risk Score, since these are already identified by the model as the highest current-risk areas.
- Treat cities flagged as Emerging Risk Zones as an early-warning list — even where absolute risk is not yet highest, a rising Incident Trend Slope indicates a location worth pre-emptive attention.
- Use the Resource Allocation Gap (Demand Z, Supply Z, Gap Score) table to guide where response capacity should be increased relative to incident demand.
- Review threat categories with the highest Weakness Score (combining SLA miss rate, escalation rate, and critical rate) as priorities for process improvement, since these represent categories the response organization currently handles least effectively.
- Monitor the 14-Day Forecast MAE over time; if backtest error grows, the forecasting approach should be reviewed before leadership continues to rely on the 14-day projection.
- Use the hour/shift/weekday patterns on the Temporal Incident Analytics page to align monitoring staff availability with actual peak incident windows.

## 16. Conclusion

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The Cyber Threat Analytics Dashboard brings together incident and response data from across the organization into a single, connected Power BI model — a two-fact, six-dimension star schema driving 41 DAX measures across seven interactive pages, a drillthrough page, and a custom tooltip page.

Its value lies in the progression it offers: from raw incident volume and severity (Security Intelligence), to geographic and temporal context (Geospatial and Temporal Analytics), to operational performance (Incident Response Analytics), to structured, multi-dimensional risk scoring (Security Risk Assessment), to a forward-looking, backtested 14-day forecast (Predictive Intelligence), and finally to a single executive summary built for rapid decision-making (Executive Intelligence Dashboard).

For a security operations audience, the dashboard supports faster risk triage and resource allocation. For management, it condenses a large, technical dataset into a small set of trustworthy, forecast-backed KPIs. For a technical reviewer or interviewer, it demonstrates practical competency in star-schema data modelling, DAX measure design, Power BI's native forecasting and Key Influencers features, forecast validation through backtesting, and multi-page interactive report design — applied to a realistic and relevant cybersecurity analytics scenario.